

Efficacy and Safety of FDC of Dapagliflozin, Glimepiride and Metformin Immediate-Release tablets in Type 2 Diabetes: A Phase 3, Active-controlled study

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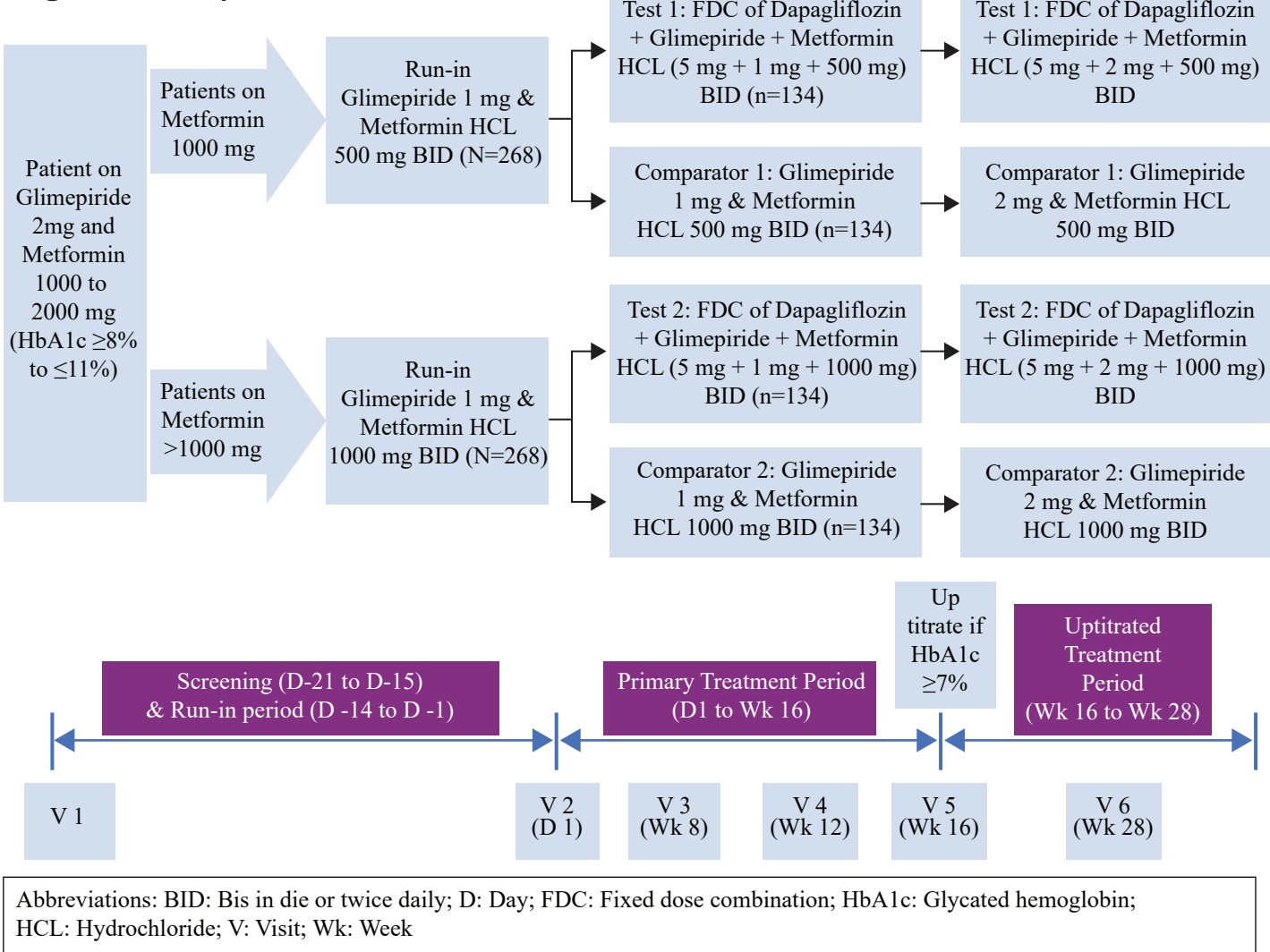
Background and Objectives

- FDC of dapagliflozin, glimepiride and metformin can offer complementary actions in simplified treatment regimen, thus improving treatment adherence and resulting in improved glycemic control in T2DM.
- The study aims to assess efficacy and safety of FDC of Dapagliflozin, Glimepiride and Metformin HCL tablets in comparison to co-administration of Metformin HCL and Glimepiride tablets in patients with T2DM.

Methodology

- **Study design:** Phase III, randomized, open-label, four-arm, multi-center, parallel-group, active controlled comparative study (CTRI/2022/03/041424).
- **Primary endpoints:**
 - Mean change in glycated hemoglobin (HbA1c) from Baseline to Week 16.
- **Secondary endpoints:**
 - Change in HbA1c at Week 12 and 28; Proportion of patients achieving HbA1c < 7.0% at the end of Week 16; Change in fasting blood glucose (FBG) and post-prandial blood glucose (PPBG) from Baseline at the end of Weeks 12, 16 and 28; Safety and tolerability of study medications.

Figure 1: Study Flow Chart



Results

- Of the 536 patients enrolled, 530 patients completed the study.
- Demography and baseline characteristics of patients were comparable between study groups (Table 1).

Table 1: Demography and Baseline Characteristics

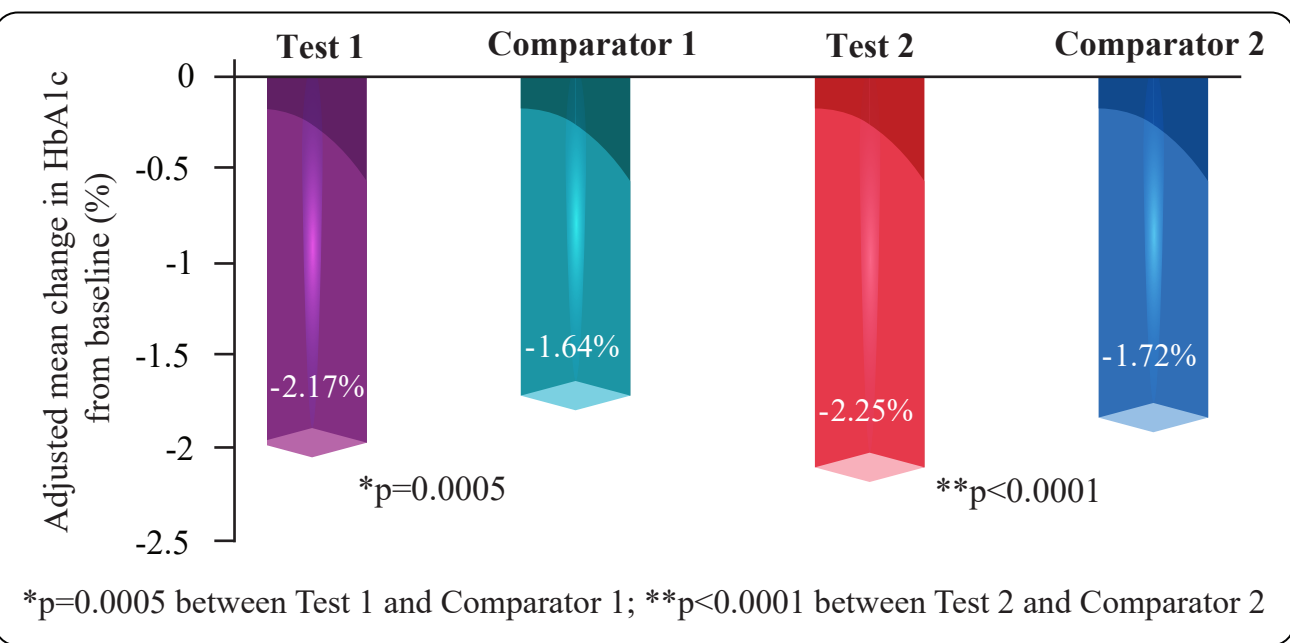
Statistic Summary	Test 1 (N = 134)	Comparator 1 (N = 134)	Test 2 (N = 134)	Comparator 2 (N=134)
Male, n(%)	73 (54.48)	74 (55.22)	78 (58.21)	76 (56.72)
Female, n(%)	61 (45.52)	60 (44.78)	56 (41.79)	58 (43.28)
Age (Years), Mean±SD	46.95±9.33	47.72±10.08	46.30±8.98	46.24±9.77
BMI (kg/m ²), Mean±SD	25.74±3.54	25.35±3.48	25.48±3.50	25.29±3.49

Primary Endpoint:

Change in HbA1c from Baseline at Week 16

- Change in HbA1c was statistically significant from Baseline to Week 16 in Test 1 vs. Comparator 1 (p=0.0005) and Test 2 vs. Comparator 2 (p<0.0001) (Figure 2).

Figure 2: Mean Change in HbA1c from Baseline to end of Week 16



Secondary Endpoints:

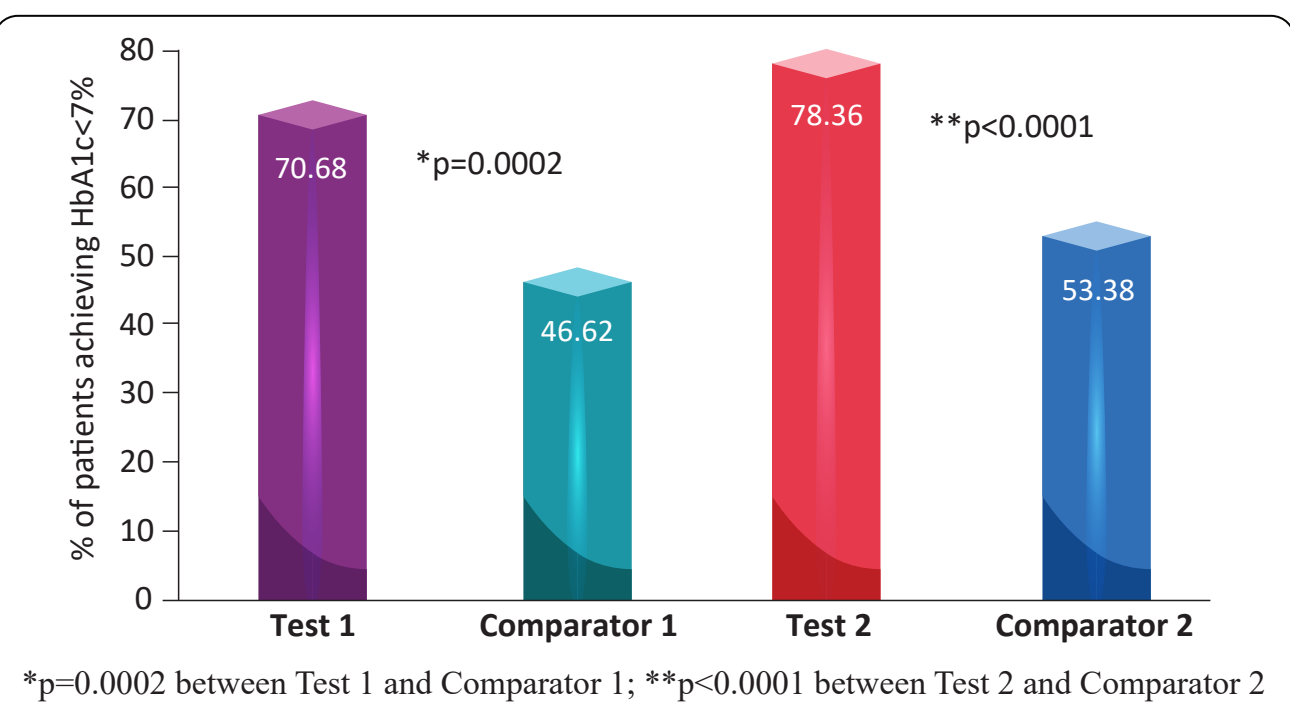
Change in HbA1c from Baseline at Weeks 12 and 28

- Reduction in HbA1c at Weeks 12 and 28 was significant from Baseline in all 4 treatment groups (p<0.0001).
- The reduction in HbA1c from Baseline to Week 12 was statistically significant in Test 2 vs. Comparator 2.

Proportion of Patients Achieving HbA1c<7%

- Proportion of patients achieving HbA1c < 7% was significantly higher in Test 1 compared to Comparator 1 at Week 16 (p=0.0002) and in Test 2 compared to Comparator 2 at Week 12 (p=0.0005) and Week 16 (p<0.0001) (Figure 3).

Figure 3: Proportion of Patients Achieving HbA1c<7% at Week 16



Change in FBG and PPBG

- Reduction in FBG and PPBG was significant in all 4 treatment groups from Baseline at Weeks 12, 16 and 28 (p<0.05).
- Reduction in FBG was significantly more in FBG in Test 2 compared to Comparator 2 at Week 12 (-28.27 vs -20.13; p=0.0169) and Week 16 (-44.23 vs -34.01; p=0.0012).
- Reduction in PPBG at Week 16 was significantly more in Test 2 vs. Comparator 2 (-69.12 vs -52.03; p=0.0074).

Safety

- **Primary treatment period:** 32 TEAEs in 26 (4.86%) patients were reported. Of 32 TEAEs: 6 in 4 (2.99%) patients in Test 1, 6 in 6 (4.51%) patients in Comparator 1, 9 in 7 (5.22%) patients in Test 2 and 11 in 9 (6.72%) patients in Comparator 2. The most commonly reported TEAEs was pyrexia, headache, and vomiting.
- **Uptitrated treatment period:** Total 4 TEAEs in 3 (1.52%) patients were reported: 1 event in 1 (2.56%) patient in Test 1 and 3 events in 2 (2.90%) patients in Comparator 1.
- **Overall safety in both periods:**
 - All events were either mild or moderate in nature.
 - No serious or severe TEAEs, or TEAEs leading to permanent discontinuation observed.
 - No hypoglycemia events were observed during the study period.
 - No patients required rescue medication or dose modification.
 - All events were resolved and reported as unlikely related to study drugs.

Discussion and Conclusion

- The reduction in HbA1c was significantly more in Test arms compared to compared to Comparator arms at Week 16.
- The proportion of patients achieving HbA1c < 7.0% were significantly higher in FDC of Dapagliflozin, Glimepiride and Metformin HCL tablets as compared to co-administration of Metformin Hydrochloride and Glimepiride tablets at the end of Week 16 demonstrating better glycaemic control.
- Overall, incidence of TEAEs were similar in Test and Comparator arm.
- To conclude, Dapagliflozin, Glimepiride and Metformin (5+1+500 BID or 5+1+1000 BID) FDC was superior in comparison to co-administration of Metformin HCL and Glimepiride without any safety concerns in T2DM patients uncontrolled on Glimepiride and Metformin combination.

Abbreviations:

BMI: Body Mass Index; FBG: Fasting blood glucose; FDC: Fixed Dose Combination; HbA1c: Glycated hemoglobin; HCL: Hydrochloride; T2DM: Type 2 diabetes mellitus; PPBG: Postprandial blood glucose; SD: Standard Deviation; TEAEs: Treatment Emergent Adverse Events.